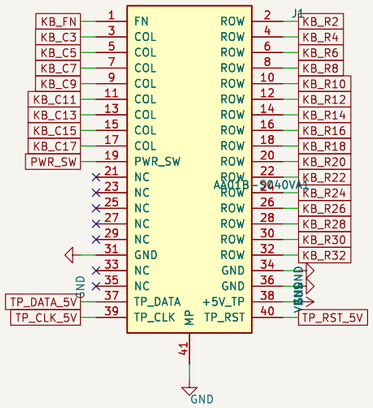
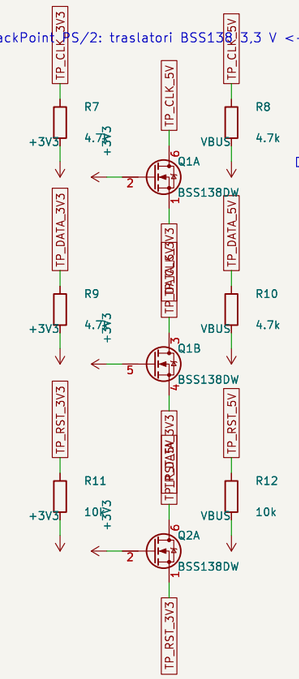
TrackPoint PS/2: traslatori BSS138 3,3 V <-> 5 V

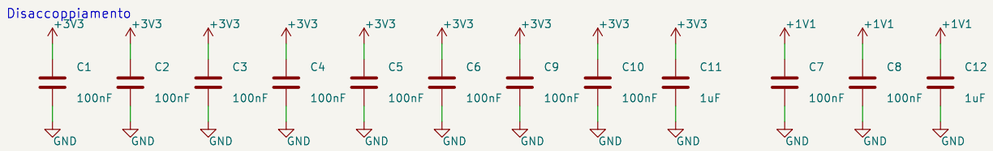
Disaccoppiamento

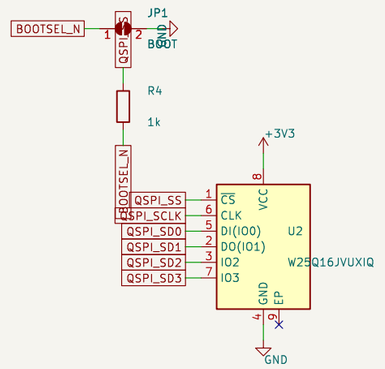
SWD e RESET

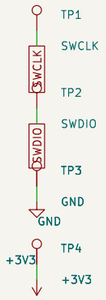
Flash QSPI 2 MB + BOOTSEL

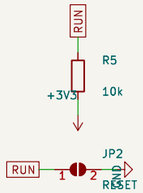








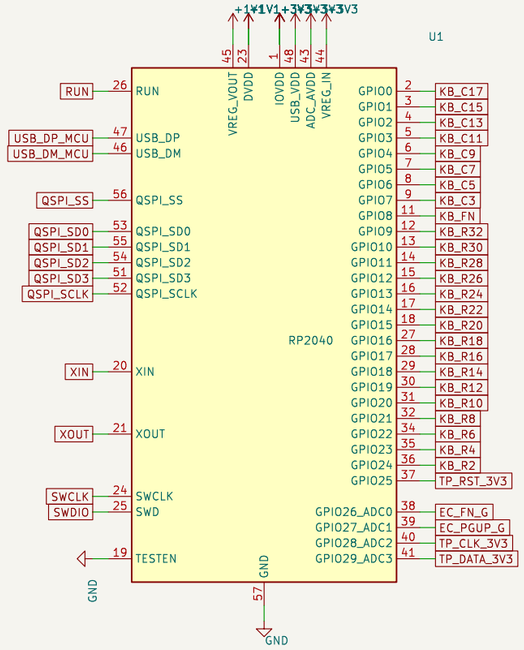


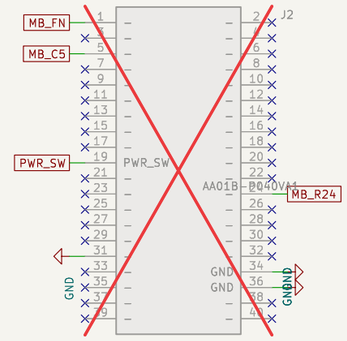


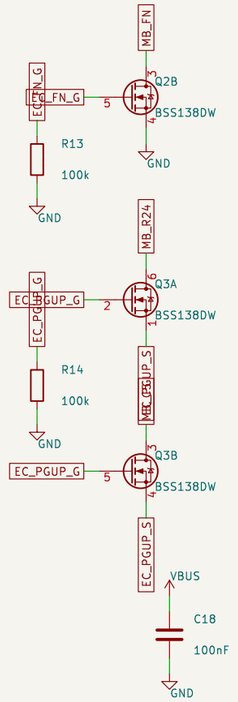
X210Ai — scheda-ponte tastiera ISO/JIS (RP2040 integrato) — rev 2.0

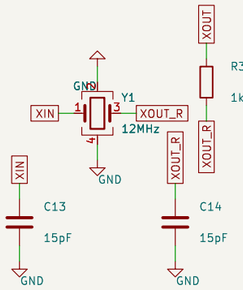
J1 sopra: tastiera. J2 sotto: BTB_R_CONN1 della scheda madre (solo pin 1, 5, 19, 24 e masse).
RP2040 legge tutta la matrice e il TrackPoint e li invia via USB (fili verso USB-CON1).
Q2b e Q3 emulano Fn+PgUp verso l'EC (ThinkLight). Pin 19 (accensione) passa diretto J1-J2.

Verso EC: Fn (J2-1) a massa, interruttore riga 24 — colonna 5 (PgUp)

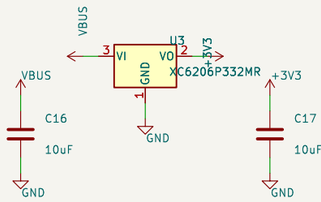




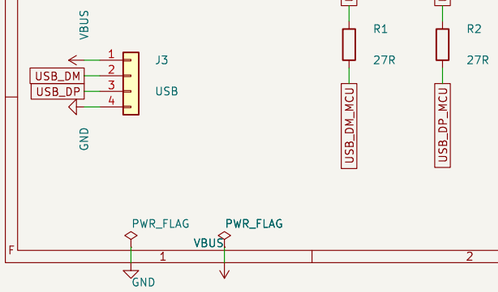




LDO 3,3 V da VBUS



USB verso USB-CON1 (4 fili)



Firmware: hektopuss ThinkPad_RP2040_Keyboard + patch X210Ai (Fn+PgUp via EC, swap Fn/Ctrl, JIS)
Scheda-ponte 4 strati: J1 sopra (tastiera), J2 sotto (scheda madre), RP2040 sotto
Basato su thedalles77 (T61 Teensy 4.0) e hektopuss (ThinkPad_RP2040_Keyboard)

Sheet: /		
File: x210ai.kicad_sch		
Title: X210Ai — scheda-ponte tastiera ISO/JIS (RP2040 integrato)		
Size: A3	Date: 2026-10-21	**Rev: 2.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1